October 30, 2026

Agenda (1/2)

- Context: the ZigBee standard and its relation to IEEE 802.15.4
- General characteristics and frequency bands
- Device roles in a ZigBee network
- Network topologies: star, tree and mesh
- Addressing and network formation
- Protocol stack and application profiles

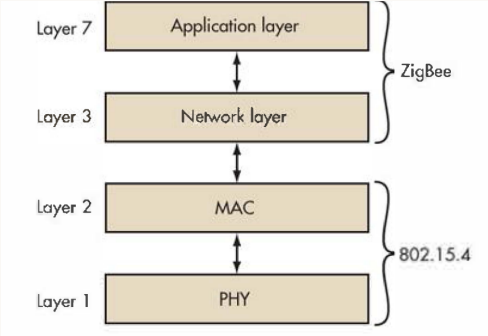
Agenda (2/2)

- Physical layer and medium access, routing, joining and security
- Energy consumption, integration in IoT architectures and use cases
- Summary

ZigBee and the IEEE 802.15.4 Standard

- ZigBee is a network specification maintained by the ZigBee Alliance (now part of the Connectivity Standards Alliance), built on the physical and medium access control layers defined by IEEE 802.15.4
- IEEE 802.15.4 defines a low-rate wireless personal area network (LR-WPAN), concerned above all with low consumption and low cost, and not with high data rates
- ZigBee adds to that base the network (NWK) and application support (APS) layers, as well as a set of standardised application profiles
- The result is a complete stack, designed for sensor and actuator networks with many nodes, battery-powered over long periods

Boundary between IEEE 802.15.4 and ZigBee



Source: Kumar, Sharma and Grewal (2014), Fig. 5.

- IEEE 802.15.4: layers 1 and 2 (PHY and MAC), that is, radio, framing, medium access and frame acknowledgement
- ZigBee: layer 3 and above (NWK, APS, ZDO and application), that is, routing, network addressing, network security and profiles
- The standard does not define multi-hop routing; ZigBee adds it
- Two products can be 802.15.4 without being ZigBee (Thread and 6LoWPAN, Lectures 09 and 10)

Terminology: IEEE 802.15.4 versus ZigBee

- Devices: the standard defines FFD and RFD (functional capability); ZigBee defines coordinator, router and end device (role in the network). The ZC and ZR are FFDs; the ZED may be an FFD or an RFD
- Coordinator: the PAN coordinator of the standard corresponds to the ZigBee Coordinator, unique per network
- Topologies: the standard provides for star and peer-to-peer; ZigBee admits star, tree and mesh, built on top of peer-to-peer
- Addresses: the standard defines the 64-bit extended address; after association, ZigBee switches to the 16-bit network address, which shortens the header and saves energy
- Joining: the new node sends a beacon request; the coordinator and the routers that allow association answer, and the node chooses the parent to attach to

General Characteristics of ZigBee (1/2)

- Very low energy consumption, with extremely low duty cycles
- Low data rate, between 20 and 250 kbps, depending on the frequency band used
- Short range, typically between 75 and 100 metres per hop
- Fast network association time, in the order of 30 milliseconds

General Characteristics of ZigBee (2/2)

- Support for small and large networks: up to 65000 devices in theory, although in practice the figure is around 240 units
- Low implementation cost, as it is an open specification protocol

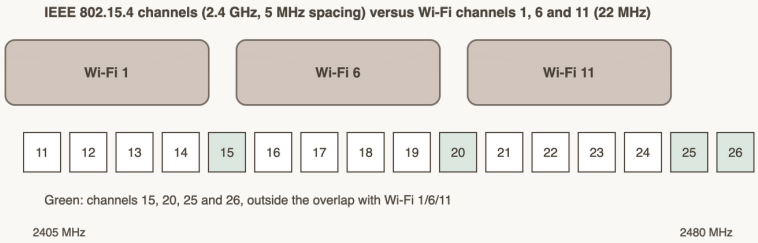
Frequency Bands and Channels

- Channel 0: the 868 MHz band, used in Europe, with a single channel available
- Channels 1 to 10: the 915 MHz band, used in the United States and Australia
- Channels 11 to 26: the 2.4 GHz band, an ISM band available worldwide, with 16 channels
- In total, the standard defines 27 channels across three frequency bands, which allows the channel choice to be adapted to the spectrum rules of each region and interference with other wireless networks to be avoided

IEEE 802.15.4 Physical Layer

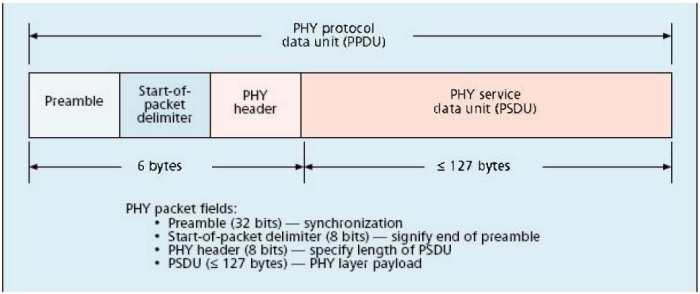
- 868 MHz (1 channel): BPSK modulation, 20 kbps; 915 MHz (10 channels): BPSK, 40 kbps
- 2.4 GHz (16 channels): O-QPSK with DSSS spreading (32 chips per 4-bit symbol), 2 Mchip/s, 250 kbps, 5 MHz channel spacing
- Physical service data unit (PSDU) of at most 127 bytes, preceded by a preamble, a start-of-frame delimiter (SFD) and a length field
- PHY services exported to the MAC: radio activation, energy detection (ED), link quality indication (LQI), clear channel assessment (CCA) and channel selection
- Typical transmit power of 0 dBm and minimum receiver sensitivity of -85 dBm at 2.4 GHz, giving indoor ranges of tens of metres

Coexistence with Wi-Fi in the 2.4 GHz Band



- A Wi-Fi channel occupies 22 MHz, equivalent to four consecutive 802.15.4 channels
- With Wi-Fi channels 1, 6 and 11, only ZigBee channels 15, 20, 25 and 26 remain free of overlap

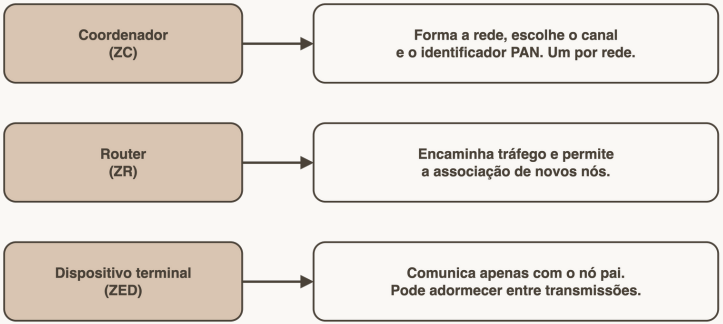
Physical Layer Packet Structure



- Synchronisation header: 32-bit preamble and 8-bit start-of-frame delimiter (SFD), followed by the 8-bit PHR with the PSDU length
- The PSDU carries up to 127 bytes: it is the complete MAC frame

Source: Kumar, Sharma and Grewal (2014), Fig. 2.

Device Types and Their Functions



Só o dispositivo terminal pode adormecer; é o único adequado a alimentação por bateria

Device Roles: the Coordinator (1/2)

- The ZigBee Coordinator (ZC) is a full function device (FFD, in IEEE 802.15.4 terminology) and there is exactly one per network
- It is responsible for starting the network, choosing the channel and the PAN identifier
- It stores the information relating to the network and to all devices that associate with it

Device Roles: the Coordinator (2/2)

- All remaining devices communicate, directly or indirectly, with the coordinator
- It may perform routing functions and act as a bridge to other networks

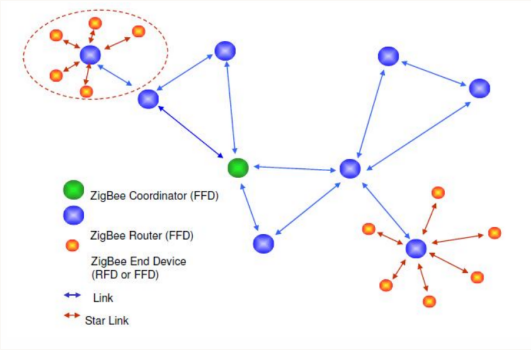
Device Roles: Router and End Device (1/2)

- The ZigBee Router (ZBR) is also an FFD, but optional in the network
- It forwards messages between nodes, acting as the backbone of the network, and extends coverage beyond the direct range of the coordinator
- It also manages the allocation and release of local addresses to the devices that associate with it

Device Roles: Router and End Device (2/2)

- The ZigBee End Device (ZED) is a reduced function device (RFD), optimised for low energy consumption
- It is the cheapest device type in the network and typically where a sensor or the equipment to be controlled is installed

ZigBee Network: Devices and Links



- Green: coordinator; blue: routers, which form the mesh among themselves (blue links)
- Orange: end devices, attached in a star to their router (clusters)
- The network is thus a mesh of routers with stars of end devices at the leaves
- Each end device depends on a single parent: that is where mesh redundancy stops protecting (lab guide)

Source: Kumar, Sharma and Grewal (2014), Fig. 6.

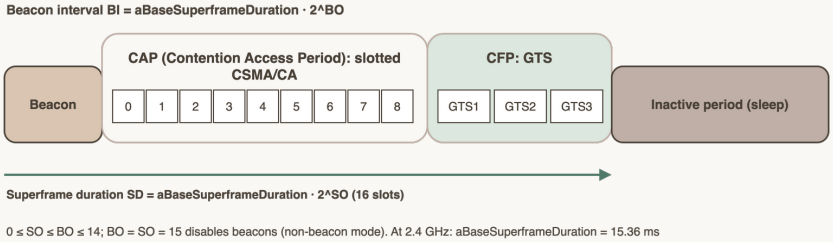
FFD, RFD and Modes of Operation (1/2)

- Full Function Device (FFD): it may take on any role in the network, including coordinator or router, and communicate with any other device
- Reduced Function Device (RFD): it has limited capabilities, communicates only with a single parent device and is intended for simple low-consumption applications
- Devices alternate between two states, active and sleeping, which saves energy when there is no data to transmit

FFD, RFD and Modes of Operation (2/2)

- The network may operate in beacon mode, with periodic synchronisation between nodes, or in non-beacon mode, without that synchronisation signal
- Traffic observed in a ZigBee network may be intermittent, repetitive or periodic, depending on the application

Medium Access: Superframe Structure

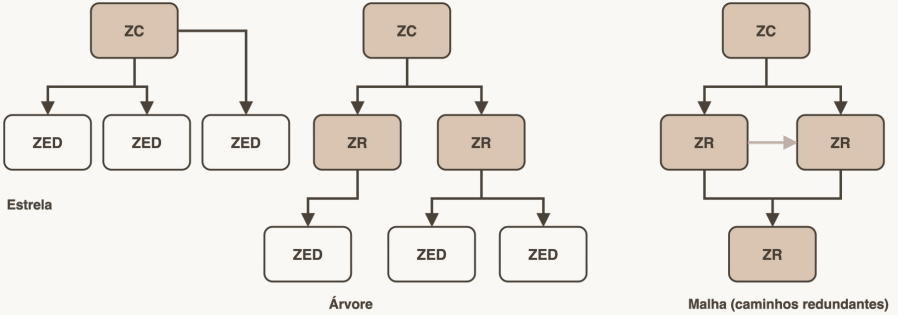


- Beacon mode: the coordinator delimits the superframe with beacons; in the CAP access is by slotted CSMA/CA, in the CFP there are guaranteed time slots (GTS) for traffic with latency requirements
- Non-beacon mode: unslotted CSMA/CA, no synchronisation; it is the mode used in ZigBee PRO mesh networks

Medium Access: CSMA/CA and Reliability

- Before transmitting, the node waits a random backoff time and assesses the channel (CCA); if busy, the backoff exponent BE grows up to macMaxBE and the attempt counter NB is incremented
- Once macMaxCSMABackoffs is exceeded, the MAC reports a channel access failure to the upper layer
- Optional acknowledgement by an ACK frame; without an ACK, the frame is retransmitted up to macMaxFrameRetries times
- There is no collision detection: the radio does not listen while transmitting, so collisions are avoided rather than detected
- Consequence for IoT: low duty cycle and short frames keep the collision probability low, but useful throughput per node falls quickly as network density grows

The Three ZigBee Topologies



Star Topology

- A single central coordinator concentrates all network communication
- All remaining devices communicate only with that coordinator, never directly with one another
- It is the simplest topology to implement and manage
- Its limitations are that the coordinator is a single point of failure and that network coverage is confined to the radio range of that single central node

Tree Topology (Cluster-Tree)

- The coordinator and the routers may emit periodic beacons, which synchronise child devices and organise medium access
- The network is organised in a hierarchical structure of parent-child relationships, starting from the coordinator
- Routers extend the reach of the network, connecting successive branches or clusters from the central node
- It is an intermediate topology between the simplicity of the star and the robustness of the mesh, but retains some dependence on the established hierarchy

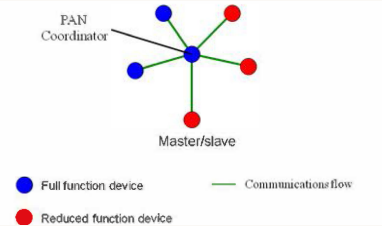
Mesh Topology

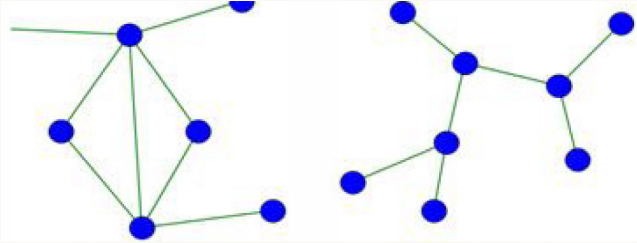
- In a mesh network, regular beacons are not permitted
- Devices communicate with one another by direct, peer-to-peer transmissions, without depending on a single central node
- Each device may forward messages on behalf of others, creating multiple possible paths between source and destination
- This path redundancy makes the network more robust to individual node failures and is the reason ZigBee is particularly suited to sensor networks and to larger-scale home automation applications

Comparison of Topologies

- Star: simplest and with lowest latency to reach the coordinator, but with no redundancy and limited coverage
- Tree: greater reach than the star, good routing predictability, but still dependent on the parent-child hierarchy
- Mesh: greater robustness and greater aggregate reach, at the cost of higher routing and network management complexity
- The choice of topology depends on the intended trade-off between simplicity, coverage, reliability and management complexity

IEEE 802.15.4 Topologies





- Star: master-slave relation; RFDs communicate only with the PAN coordinator
- Peer-to-peer: FFDs only, in direct communication, without beacons; allows a free point-to-point or a cluster-tree structure

Source: Kumar, Sharma and Grewal (2014), Figs. 3 and 4.

Device Addressing (1/2)

- Two or more devices communicating on the same physical channel constitute a WPAN, which always includes at least one PAN coordinator
- Each independent PAN chooses a unique identifier, the PAN ID
- Every device has a 64-bit extended address, inherited from IEEE 802.15.4, which may be used for direct communication within the PAN

Device Addressing (2/2)

- On association, the coordinator assigns each device a 16-bit short address, more efficient for routine use in the network
- Up to 256 sub-addresses may further be allocated to subunits of the same device

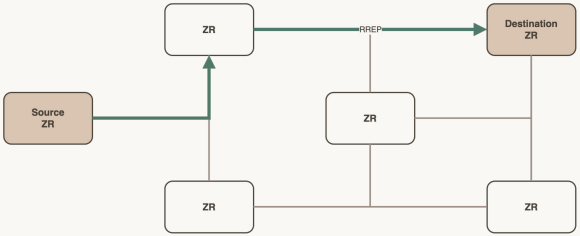
Hierarchical Addressing and Tree Routing

- Network parameters: C_m (maximum children), R_m (maximum children that are routers) and L_m (maximum depth)
- Each router at depth d receives a block of addresses; the range reserved for each sub-branch is

$$C_{skip}(d) = \begin{cases} 1 + C_m(L_m - d - 1), & R_m = 1 \\ \dfrac{1 + C_m - R_m - C_m R_m^{L_m - d - 1}}{1 - R_m}, & R_m > 1 \end{cases}$$

- The coordinator has address 0x0000; the address of the n-th router child is $A_{parent} + C_{skip}(d)(n-1) + 1$
- With blocks assigned, a router decides without tables: if the destination belongs to its block, it goes down to the matching child; otherwise it goes up to its parent

Mesh Routing: Route Discovery (AODV)



- The source broadcasts a Route Request (RREQ); each router accumulates the link cost and rebroadcasts
- The destination answers by unicast with a Route Reply (RREP) along the lowest-cost path

Routing: Tables, Cost and Broadcast

- Each router keeps a routing table (destination, next hop, state) and a route discovery table (RREQ identifier, accumulated cost)
- The cost of each link (1 to 7) derives from measured quality (LQI, loss rate); the chosen path minimises the sum of costs, not the number of hops
- Many-to-one: a concentrator (for example the gateway) announces itself by broadcast and all nodes learn a single route to it; it is the typical data collection pattern in IoT
- Broadcasts: rebroadcast by all routers, with a broadcast transaction table (BTT) and a maximum radius; a rebroadcast overheard from a neighbour acts as a passive acknowledgement
- When a route fails, the source repeats the discovery; the mesh recovers without intervention, at the price of latency and control traffic

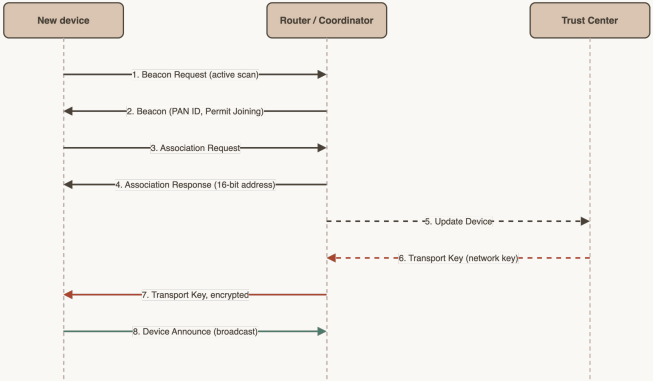
Network Formation and Associated Mechanisms (1/2)

- Stochastic addressing: each device is assigned a random address, which is then announced, with a conflict resolution mechanism in place; parent nodes need not maintain a table of already assigned addresses
- Link management: each node records the quality of the link to each neighbour, using it as a routing cost
- Frequency agility: a node suffering interference reports it to a channel manager, which may select another channel for the whole network

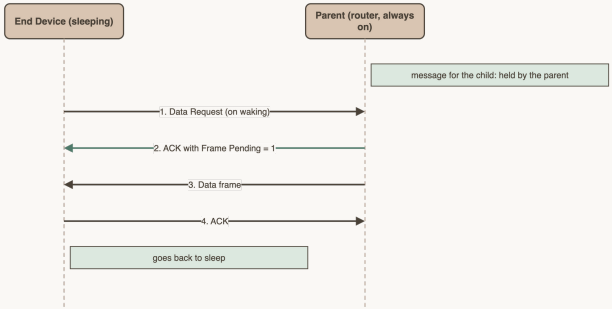
Network Formation and Associated Mechanisms (2/2)

- Asymmetric links: each node has its own transmit power and sensitivity, so paths between two nodes may have different quality in each direction
- Power management: coordinators and routers are typically mains-powered, while end devices run on batteries

Joining a Device to the Network

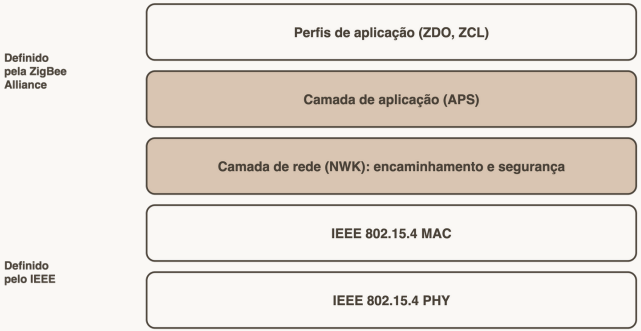


Indirect Delivery to Sleeping End Devices



- The end device never receives spontaneously: the parent holds the messages and delivers them when the child asks for them
- The poll period sets the trade-off between delivery latency and consumption

Protocol Stack and Responsibilities



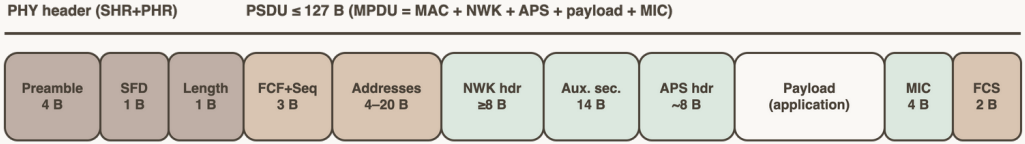
ZigBee Protocol Stack (1/2)

- Physical layer (PHY) and medium access control layer (MAC), inherited directly from IEEE 802.15.4
- Network layer (NWK): responsible for topology, routing and address assignment
- Application support layer (APS): it bridges the network layer and the applications, managing bindings between devices

ZigBee Protocol Stack (2/2)

- ZigBee Device Object (ZDO): it defines the role of the device in the network (coordinator, router or end device), initiates and responds to binding requests, and establishes security relationships between devices
- Application layer: it hosts the application profiles and the objects representing the concrete devices

Frame Format and Encapsulation

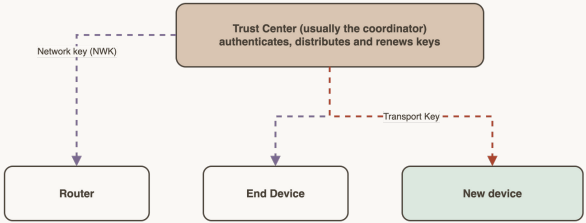


Grey: PHY. Sand: MAC (IEEE 802.15.4). Green: ZigBee layers (NWK, APS and AES-CCM* security)

Application payload at most ≈ 70–80 B per frame

- Each layer adds its own header; the 127-byte PHY limit reduces the payload to a few tens of bytes
- Larger messages require fragmentation at the APS

Security (1/2)



Network key (128 bit): encrypts and authenticates all NWK frames; shared by all nodes

Link key (APS): between two nodes or node/Trust Center; Install Code in ZigBee 3.0

AES-128 in CCM* mode: confidentiality + MIC (32 bits); frame counter against replay

Security (2/2)

- AES-128 symmetric encryption in CCM* mode: optional confidentiality, plus integrity and authentication through a MIC appended to the frame
- Two key layers: the network key protects NWK traffic; link keys protect the APS between two nodes
- The Trust Center, usually the coordinator, authenticates new nodes and distributes or renews the network key
- Frame counters protect against replay of captured messages
- Historical weak point: publicly known default link keys allow the network key to be captured during joining; ZigBee 3.0 Install Codes assign a unique key per device

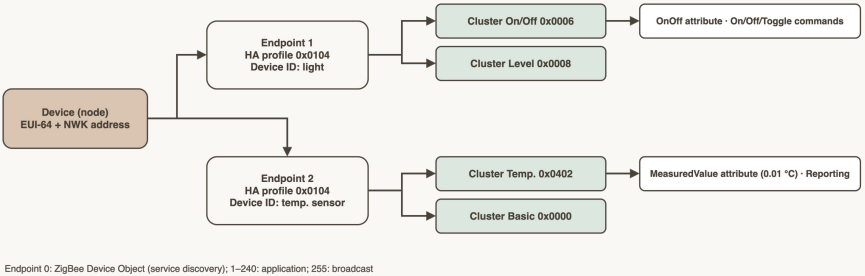
Application Profiles and ZigBee Device ID (1/2)

- Application profiles define a common language for data exchange, specifying the services offered and ensuring interoperability between devices from different manufacturers
- Public profiles are made available by the ZigBee Alliance, contain standardised device descriptions and a unique identifier licensed by the Alliance
- Manufacturer-specific profiles are intended for proprietary applications, developed privately; they must nonetheless use a profile identifier assigned by the Alliance, and commercial products implementing them must undergo network capability testing

Application Profiles and ZigBee Device ID (2/2)

- Each endpoint of a device includes a profile identifier and a device identifier (device ID), representing something physical and concrete: a lamp, a temperature sensor, a thermostat, for instance
- Device IDs lie in the range between 0x0000 and 0xFFFF and their meaning depends on the profile to which they belong; the Home Automation Profile is an example of a profile defining its own set of device IDs

Data Model: Endpoint, Cluster and Attribute



- The ZigBee Cluster Library (ZCL) standardises clusters, attributes and commands, including periodic attribute reporting
- A binding links an output cluster on one node (for example a switch) to an input cluster on another (a lamp)

Energy Consumption (1/2)

- ZigBee was designed from the outset for very low duty cycles, with devices spending most of their time in sleep mode
- End devices wake only periodically to check with the parent device whether there are pending messages, which allows them to run for years on a single battery

Energy Consumption (2/2)

- Coordinators and routers must remain permanently active in order to forward traffic, and are therefore normally mains-powered
- The fast network association time, in the order of 30 milliseconds, also contributes to reducing consumption, since devices need not remain active for long in order to join the network

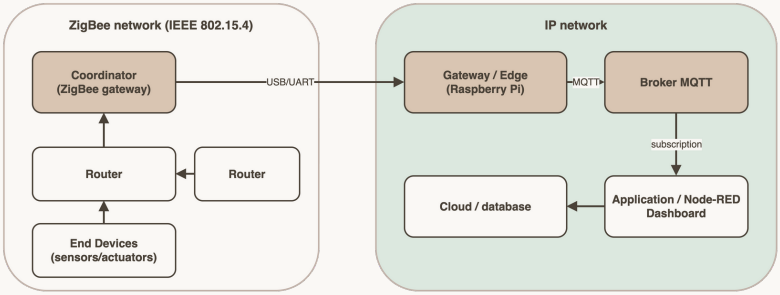
Duty Cycle and End Device Lifetime

- Average current: $I_{avg} = d \cdot I_{active} + (1 - d) \cdot I_{sleep}$, with d the duty cycle; lifetime $\approx C / I_{avg}$
- Order of magnitude for a sensor reporting every 10 s, 5 ms active at 25 mA and 1 μA asleep: $d = 5 \times 10^{-4}$ and $I_{avg} \approx 13.5$ μA
- With a CR2032 cell of ≈ 225 mAh, the theoretical lifetime is about 1.9 years; polls to the parent and cell self-discharge change this figure
- Reducing the duty cycle (reporting only on event or threshold) is the measure with the greatest effect on lifetime

Typical Use Cases

- Home automation: lighting, thermostats, locks and other devices that need to communicate with one another and with a central control hub
- Wireless sensor networks: environmental, industrial and agricultural monitoring, with many nodes spread over a wide area
- Smart buildings and commercial lighting, where the mesh topology allows large areas to be covered without a single central access point
- In all these scenarios, the decisive factor is not data rate, but the energy autonomy of the devices and the scalability of the network to hundreds of nodes

Integration in an IoT Architecture



ZigBee ⇄ IP translation at the gateway (Lecture 13: Zigbee2MQTT)

– ZigBee occupies the access layer: it is not IP-routable, so the gateway translates to MQTT or HTTP

EXAMPLE | Home Automation: Lamps and Switches

- Commercial systems such as Philips Hue and IKEA TRÅDFRI use ZigBee between the hub and the lamps
- Roles: mains-powered lamps are routers and form the mesh; motion sensors, remotes and buttons are battery-powered end devices
- Topology: mesh, which extends the hub's reach to the whole house without cabling
- Interoperability: the Home Automation profile (and later ZigBee 3.0) allows mixing manufacturers on the same hub

EXAMPLE | Smart Metering

- The Smart Energy profile is used in the home area networks (HAN) of smart meters, for example in the United Kingdom (SMETS2), between the meter, the communications hub and the in-home display
- Requirements: strong security (keys and Trust Center), low data rate and a service life of years without maintenance
- Roles: the electricity meter is a router; the battery-powered home display is an end device
- The link to the operator (WAN) uses another technology; ZigBee covers only the last metre

EXAMPLE | Smart Buildings and Lighting

- Control of lighting, HVAC and occupancy in offices, hotels and hospitals, with hundreds of nodes per building
- Mesh topology with many routers: the failure of one luminaire does not isolate the rest of the network
- ZigBee Green Power: battery-less switches, powered by the energy of the press itself, send short frames that a proxy forwards into the network
- Connection to the building management system through an IP gateway

EXAMPLE | Agriculture, Industry and Healthcare

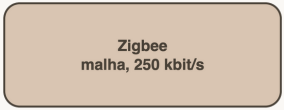
- Agriculture and environment: battery-powered soil moisture and temperature sensors, collected periodically by a sink node; the many-to-one pattern and low duty cycle are decisive
- Industry: monitoring of vibration, temperature and consumption in places without cable infrastructure, with path redundancy
- Healthcare: the ZigBee Health Care Profile, based on the ISO/IEEE 11073 standards, defines data exchange for personal devices such as scales and blood pressure monitors
- Common criterion: many nodes, short messages, long lifetime and tolerance to failures of individual nodes

EXAMPLE | ZigBee Application Families

- Building Automation: monitoring and control of commercial facilities
- Home Automation and Light Link: smart homes and LED lighting
- Smart Energy: home energy monitoring and smart metering
- Health Care: medical and fitness monitoring
- RF4CE (Remote Control) and Input Devices: remote controls and peripherals such as keyboards, mice and touchpads
- Retail Services, Telecom Services and large mesh networks
- Each family corresponds to an application profile with its own device IDs and clusters

List source: Kumar, Sharma and Grewal (2014).

ZigBee Compared with Other Short-Range Technologies



Consumo muito baixo
Alcance alargado por salto



Consumo baixo
Sem encaminhamento entre nós



Consumo elevado
Infraestrutura existente

Advantages and Limitations (1/2)

- Advantages: low energy consumption; security and application support services built on IEEE 802.15.4; low cost of modules and chips; mesh topology, with no need for a central hub; high reliability even in adverse conditions
- Limitations: short range, which may be insufficient in larger buildings; low data rate, poorly suited to bulk transfers; smaller market adoption than other IoT protocols, with an impact on interoperability between manufacturers; security mechanisms less robust than those of more recent alternatives

Advantages and Limitations (2/2)

- Compared with technologies such as Bluetooth Low Energy, ZigBee is distinguished by its native mesh topology and by an established presence in home automation applications, although BLE has been gaining ground in energy consumption and in presence on mobile devices

Evolution: ZigBee 3.0 and Matter

- ZigBee PRO (2007) and revision R21 consolidated routing, security and stochastic addressing
- ZigBee 3.0 (2016) unifies the earlier profiles (Home Automation, Light Link, among others) into a single specification
- The ZigBee Alliance became the Connectivity Standards Alliance (CSA) in 2021, which also maintains Matter
- Matter defines the application layer over IP (Wi-Fi, Ethernet and Thread, which is also based on IEEE 802.15.4); existing ZigBee devices are integrated through bridges

Summary (1/2)

- ZigBee is a low-power mesh network stack, built on IEEE 802.15.4, aimed at sensor and actuator networks with many nodes
- It does not seek high data rates or long range; it seeks energy efficiency, low cost and scalability to hundreds of devices
- The coordinator, router and end device roles, combined with the star, tree and mesh topologies, give flexibility to adapt the network to different deployment scenarios

Summary (2/2)

- Application profiles ensure that devices from different manufacturers can interoperate, which is essential in home automation applications
- It coexists with other short-range technologies, such as Bluetooth Low Energy, the choice resting on the specific nature of each application

Sources (1/2)

Gonçalves, P. *Apresent6 — Zigbee*, TCIoT, previous editions. Basis for the general characteristics, device roles, topologies, addressing and application profiles.

IEEE Std 802.15.4-2020. Physical layer, superframe and CSMA/CA, on which ZigBee rests.

Connectivity Standards Alliance. *ZigBee Specification* (document 05-3474) and *ZigBee Cluster Library*: routing, addressing, security and data model.

Sources (2/2)

Kumar, A., Sharma, A. and Grewal, K. (2014). Resolving the paradox between IEEE 802.15.4 and Zigbee. *ICROIT 2014*, pp. 484–486. IEEE.

Ramya, C. M. *et al.* (2015). A Survey of ZigBee Wireless Sensor Network Technology: Topology, Applications and Challenges. *IJCA* 130(9). Available in `Leituras/`.

Kaushal, K. *et al.* ZigBee based Wireless Sensor Networks. *IJCSIT*. Available in `Leituras/`.

Farahani, S. (2008). *ZigBee Wireless Networks and Transceivers*. Newnes. Gislason, D. (2008). *Zigbee Wireless Networking*. Newnes.

Use cases: public documentation from manufacturers (Philips Hue, IKEA TRÅDFRI) and from the CSA (Smart Energy, Green Power and Health Care profiles).

Thank you.

Questions and comments